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Bean Counter Bakery Cafe: What is it?

In 1988, Alice Lombardi started Bean Counter Bakery Café out of her licensed home kitchen in Worcester. After years of community support, she was able to open her first café on Highland Street in Worcester, Massachusetts in 2001, and then it expanded to another location on Grove Street in May of 2020. Over the last 20 years, Bean Counter Bakery Café was able to grow from a one-person operation, to a multi location bakery business (Bean Counter Bakery, 2026).

Bean Counter Bakery Cafe typically focuses on local bakery and cafe needs, as well as the creation of specialty coffees, baked goods, and breakfast/lunch items. They are a local company with two locations surrounding WPI (Bean Counter Bakery, 2026). Their store includes the store management, the front-of-house staff, and the kitchen and bakery staff.

Issues with Bean Counter Bakery Cafe

As Bean Counter Bakery Cafe typically works around a local community, there are many familiar customers and the community is very loyal. However it took a long time, around 13 years, to save up enough to afford its own location (Bean Counter Bakery, 2026). Our Business Intelligence team was tasked with analyzing Dunkin Donut’s Stores datasets to see how insights in similar large cafes can be applied to Bean Counter Bakery Cafe. Our team analyzed sales, location, revenue, and foot traffic by different stores, as well as two interactive dashboards (a Sales Dashboard and a Digital Capacity Dashboard) and implementation recommendations for Bean Counter Bakery Cafe.

Expected value of BI for client company:

Business intelligence (BI) is the broad term for the technologies, applications, and processes used for gathering, cleaning, storing, analyzing, and visualizing data to help users make better decisions. BI can help businesses with problems involving data management, data reporting, performance and operational inefficiencies, sales challenges, and risk management.

BI Framework for Bean Counter:

Data Sources:

- Sales records
- Product Inventory Data
- Foot Traffic measured by time

Data Layer:

- Converted dataset from json to csv and cleaned dataset

Analytics

- Revenue and Sales Data
- KPI calculations and visualizations
- Product transaction mix(food and beverage)

Dashboards/Presentation

- Sales dashboard
- Digital Capacity Dashboard

Case Study - McDonalds

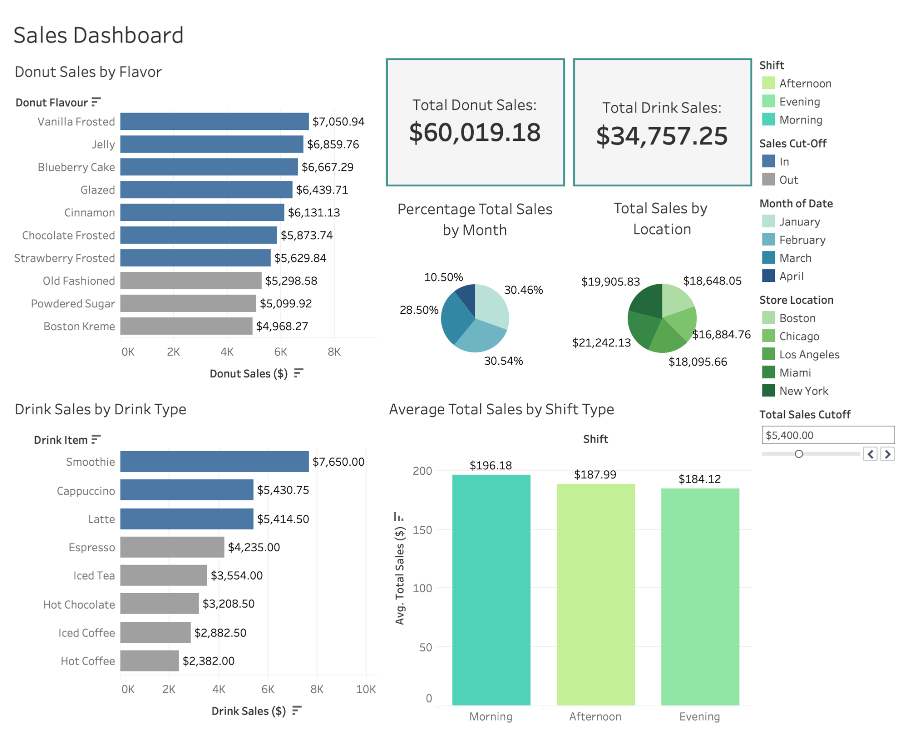
Using Enterprise Resource Planning(ERP) and BI Deployment, McDonalds was able to expose data quality issues, fill significant performance and scalability deficiencies, and resolve performance issues that had been impacting user adoption (KPI Partners New Team, 2026). McDonalds is an example of a restaurant chain that made great improvements through the use of BI, and Bean Counter Cafe could utilize the same strategies to grow their business as well through the use of dashboards to help analyze past trends, and make informed decisions for the future of their company.

Dashboard One: Sales Dashboard

Intended Users: Sales executives and managers

KPIs we looked at:

- Most sales for specific time periods(morning, afternoon, night)
- Total Drink Sales
- Total Donut Sales
- Sales by Location
- Daily/Monthly Revenue trend



This dashboard looks at the donut sales by flavor, drink sales by drink type, percentage of total sales by month, total sales by location, and average total sales by shift type (morning, afternoon, and evening). One of the filters we added to this dashboard was filtering by month. For example, if you click on a specific month in the pie chart for “Percentage of Total Sales by

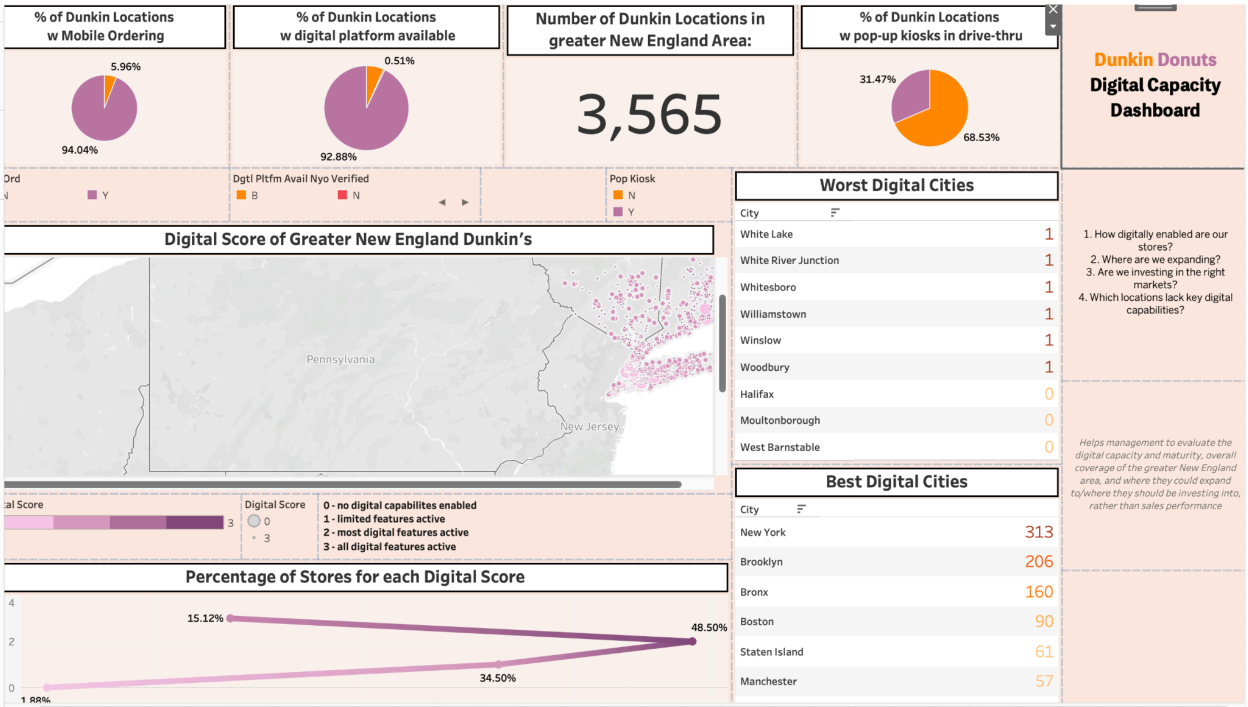
Month", the dashboard only displays the data from that month adjusting all the charts, numbers, and percentages to fit those data points. It is the same for sales by location, if you click on a slice for the "Total Sales by Location", then the whole dashboard adjusts for that specific location. Finally the filters also work for shift type, if you click on a specific bar on the "Average Total Sales by Shift Type", the data only displays data from that time of day. There is also a user-input slider for total sales cut-off. This can be used to set the limit for what counts as being "profitable", and the user can change the cut-off point if there is a specific threshold that the company is looking at or wants to look at.

Dashboard Two: Digital Capacity Dashboard

Intended users: Sales executives and managers

KPIs we looked at:

- Amount of Digital Features by location
- Potential Market demand by region
- Sales Performance vs Digital investment
- Locations without Digital Features



This Dashboard looks at the digital features, more specifically the Dunkin Donuts locations with mobile ordering, digital platforms, and with pop up kiosks in the drive through. Most locations had mobile ordering and digital platforms but not pop-up kiosks in the drive through. From the analysis, we looked specifically into the New England regions and analyzed using a metric created to test their digital capacity called "digital score" (see dashboard for more explanation on what each score values means). Utilizing the digital score we found the two best digital cities are New York City and Brooklyn, while the worst two digital cities are White Lake and White River Junction. We also looked at the percentage of stores for each digital score, where around 60-65% of stores have at least two digital features. Some of the features on this dashboard include a filter by region, as well as a filter for each digital score.

Discussion: Managerial Aspects

Bean Counter should choose performance indicators that align with their strategic goals such as increasing sales, improving customer satisfaction, and expansion as shown in our proposed BI solutions.

Potential KPIs include

- Daily revenue
- Sales by product category
- Peak-hours sales trends
- Customer return rate
- Customer satisfaction
- Mobile app usage by Location
- Average foot traffic by Location

Discussion: Technical Aspects

Bean Counter relies on multiple data sources such as sales records, inventory records, supplier invoices, and online ordering platforms. With this there could be inconsistency with product naming, duplicate records, inaccurate inventory counts, or delays with updating information. Implementing naming conventions and using software, such as Tableau, to help clean and validate data would be necessary. The software could help flag unusual entries (with potentially incorrect data) or duplicate information. Additionally, integration of databases such as the sales and inventory dashboards would help reduce discrepancies within the data.

Discussion: Ethical and Legal Aspects

If Bean Counter Café opens in Europe, it must comply with the General Data Protection Regulation(GDPR). Article 44 requires that any transfer of personal data outside the EU maintains equivalent data protection standards. Article 21 gives customers the right to object to their personal data being used for direct marketing, including promotional communications. One solution to this could be implementing a data protection officer, as this person can be the one to mediate between the companies' regulatory authorities, and relevant data subjects. Another solution could be implementing a privacy policy for people trying to understand how data is being stored and collected, but one has to make sure the privacy policy is read and understood by relevant subjects.

Other Processes and Artififacts

Social Media Analytics can help us identify what processes work for the company, getting customer thoughts on different processes, and determine the operations that can be improved, which can help optimize the dashboards as well as other business operations. Some important techniques that can be useful are Sentiment Analysis and Social Network Analysis. Sentiment Analysis is a technique that can analyze the emotion of a certain text, whether it is positive, negative, or neutral. This is important because having a software that can program human emotion makes it easier to decipher the analysis. Quantifying the analysis goes hand-in-hand with other potential calculations. Another technique is Social Network Analysis, which analyzes the structure of a social network. Some applications include examining the relationships among the people, identifying the patterns of information flow, and identifying key opinion leaders. This is important because you can see the most popular customers, the most popular features being used (different digital features), this can also help identify trends, which is important if the company is looking to expand the stores to different locations.

Summary and Conclusion

In the sales dashboard, some insights to summarize the findings include total sales was higher for the donut than drink sales. We also noticed a combination of higher sales in winter months and during morning shift hours, and the two locations with the highest sales are in LA and NY. In the Digital Capacity dashboard, we noticed there were a lot of locations that have both mobile ordering and digital platforms. More than 65% of locations have at least 1 digital platform. There are less locations with pop-up kiosks in drive-thru. We recommend locations to invest in more technology, and to innovate and expand to locations with this higher digital capacity. The data supports the stores with digital innovation with one stand out location being the New York regions, as they have robust digital capacity and overall high sales.

References

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